# THREE BODY PROBLEM IN CELESTIAL MECHANICS

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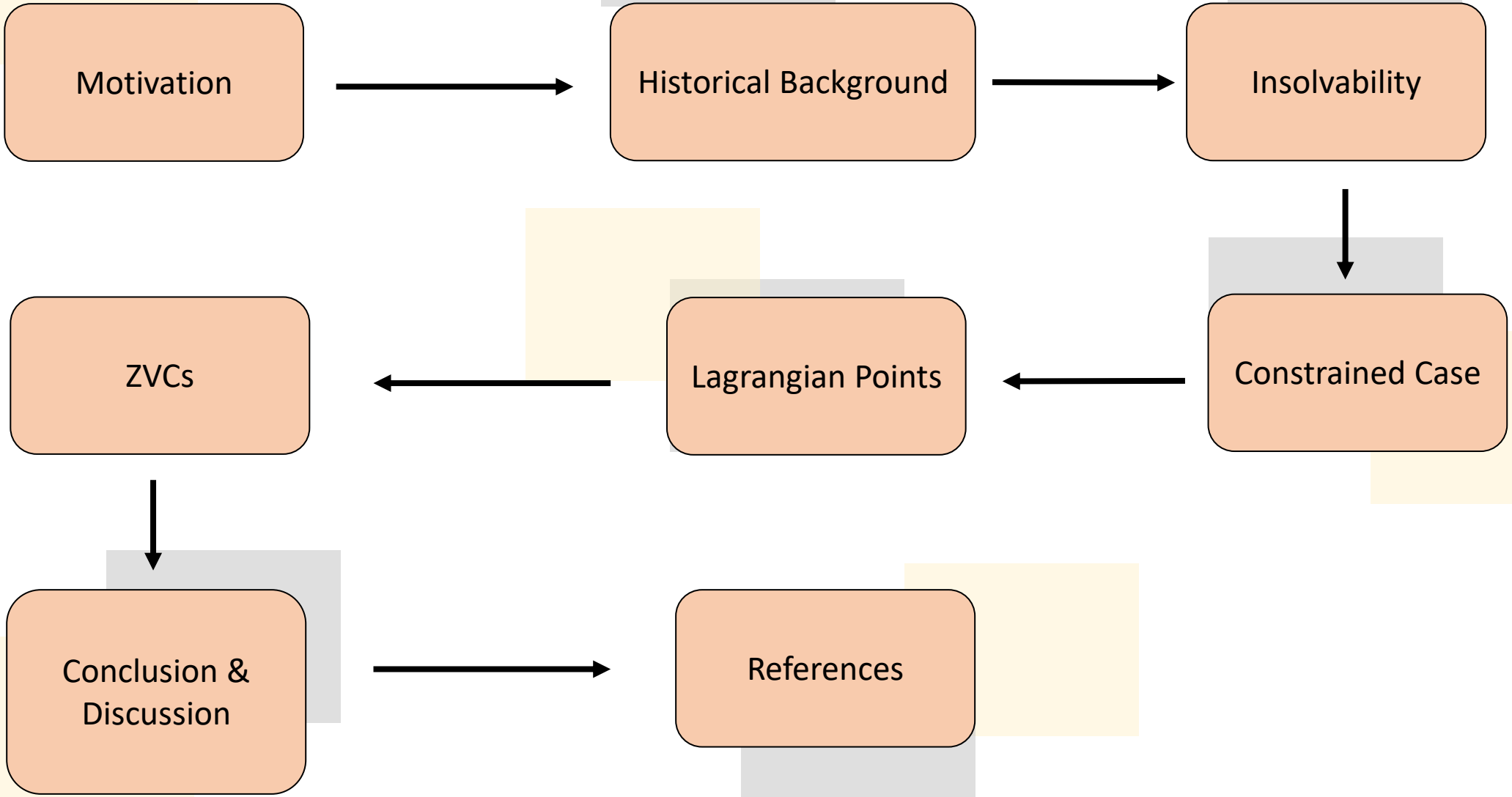
# PRESENTATION MAP



# MOTIVATION

Three Body problem is the understanding of motion of three bodies under the influence of each other gravitational force.

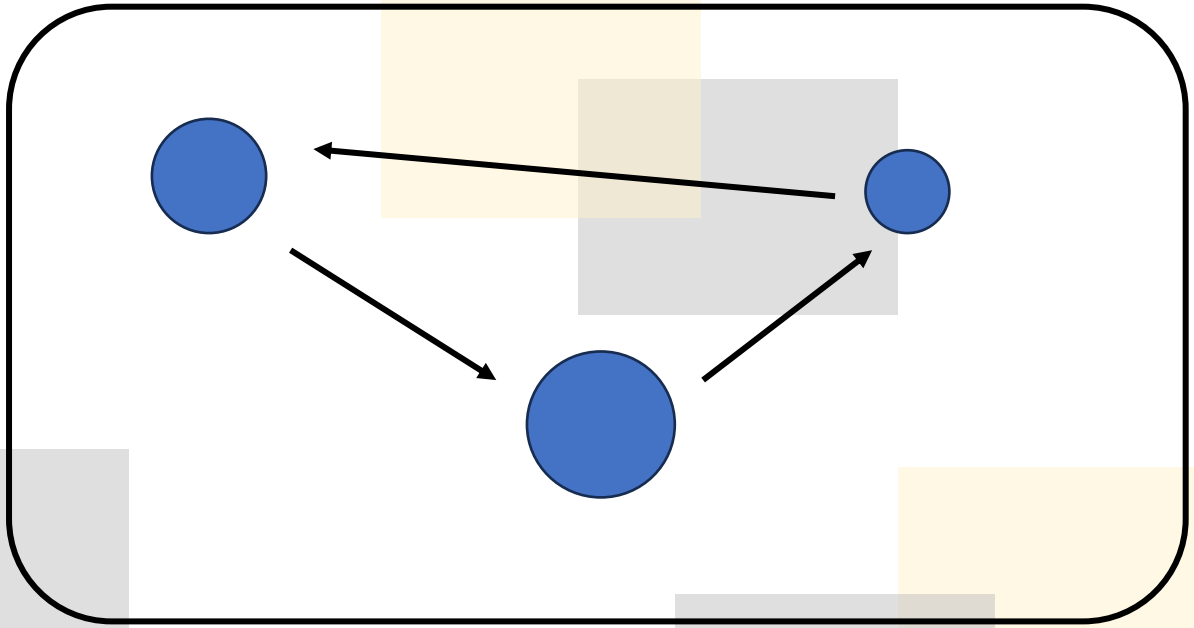


Fig 1. Three Body Problem

This problem is unsolvable. Why?.....

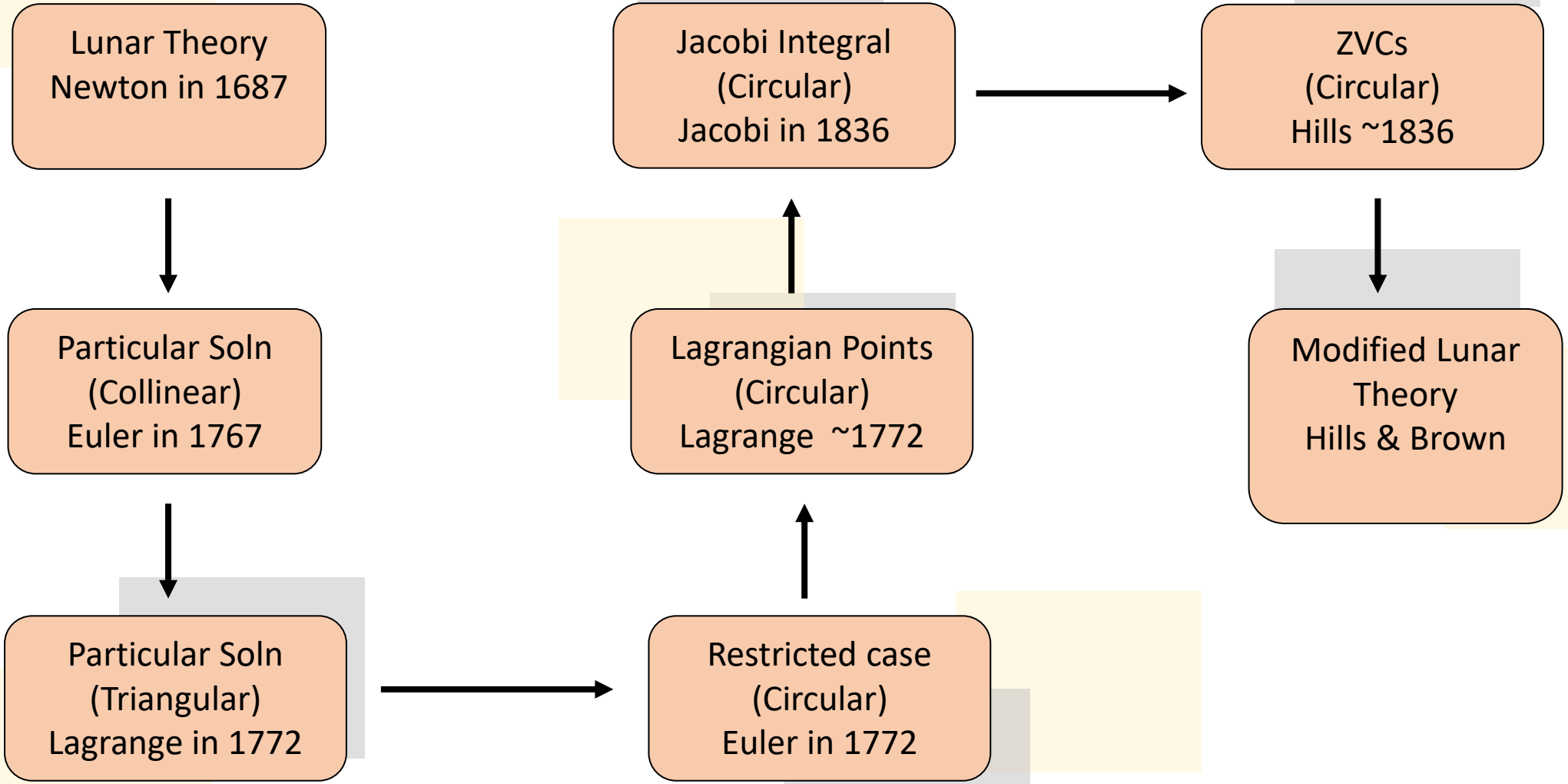
This problem was very difficult for newton himself that it was stated,

"…….the consequence of which includes a headache."[1]

No finite or linear way of solving the problem!

This problem was studied so much that more than 800 memoirs were to published from 1750 to 1917. [2]

# HISTORICAL BACKGROUND



# INSOLVABILITY

1. $\ddot{R}_1 = G\frac{M_2}{r^3{}_{12}}\overrightarrow{r_{12}} + G\frac{M_3}{r^3{}_{13}}\overrightarrow{r_{13}}$
2. $\ddot{R}_2 = G\frac{M_1}{r^3{}_{21}}\overrightarrow{r_{21}} + G\frac{M_3}{r^3{}_{23}}\overrightarrow{r_{23}}$
3. $\ddot{R}_3 = G\frac{M_2}{r^3{}_{32}}\overrightarrow{r_{31}} + G\frac{M_1}{r^3{}_{32}}\overrightarrow{r_{32}}$

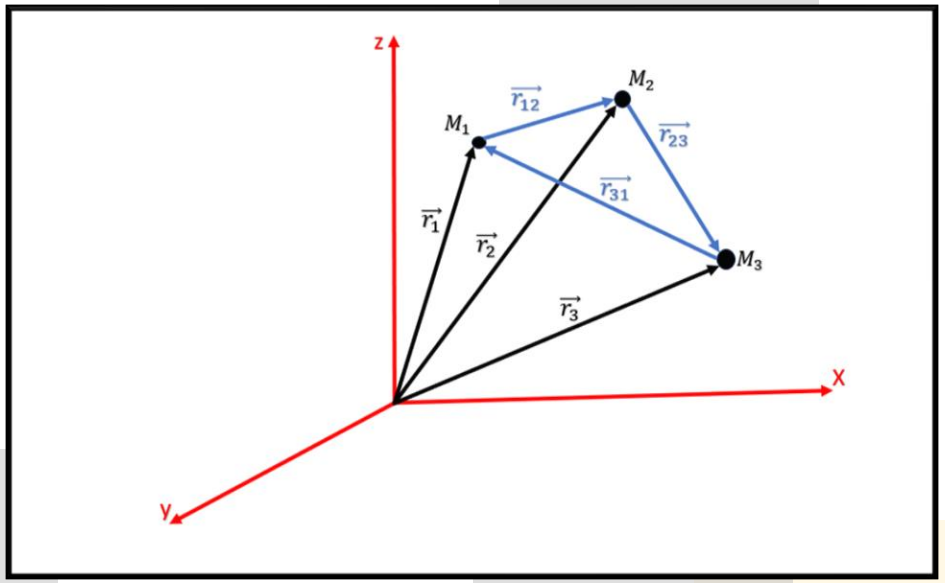



Fig 2. Three Body Problem

Here, we have 3, 2nd order D.E and in axis form, we have 9, 2nd D.E.

In total we have 19, 1st order D.E.

# Integrals of Motion

Along Conserved Linear Momentum → 6, integral of motion

Along Conserved Angular Momentum → 3, integral of motion

Along Conserved Total Energy → 1, integral of motion

Along orientation of bodies → 2, integral of motion

In 1887, Bruns mathematically proved that no other integral of motions exists for the problem
Hence, Still 6 equations left to solve, which makes the problem unsolvable.

# CONSTRAINED CASE

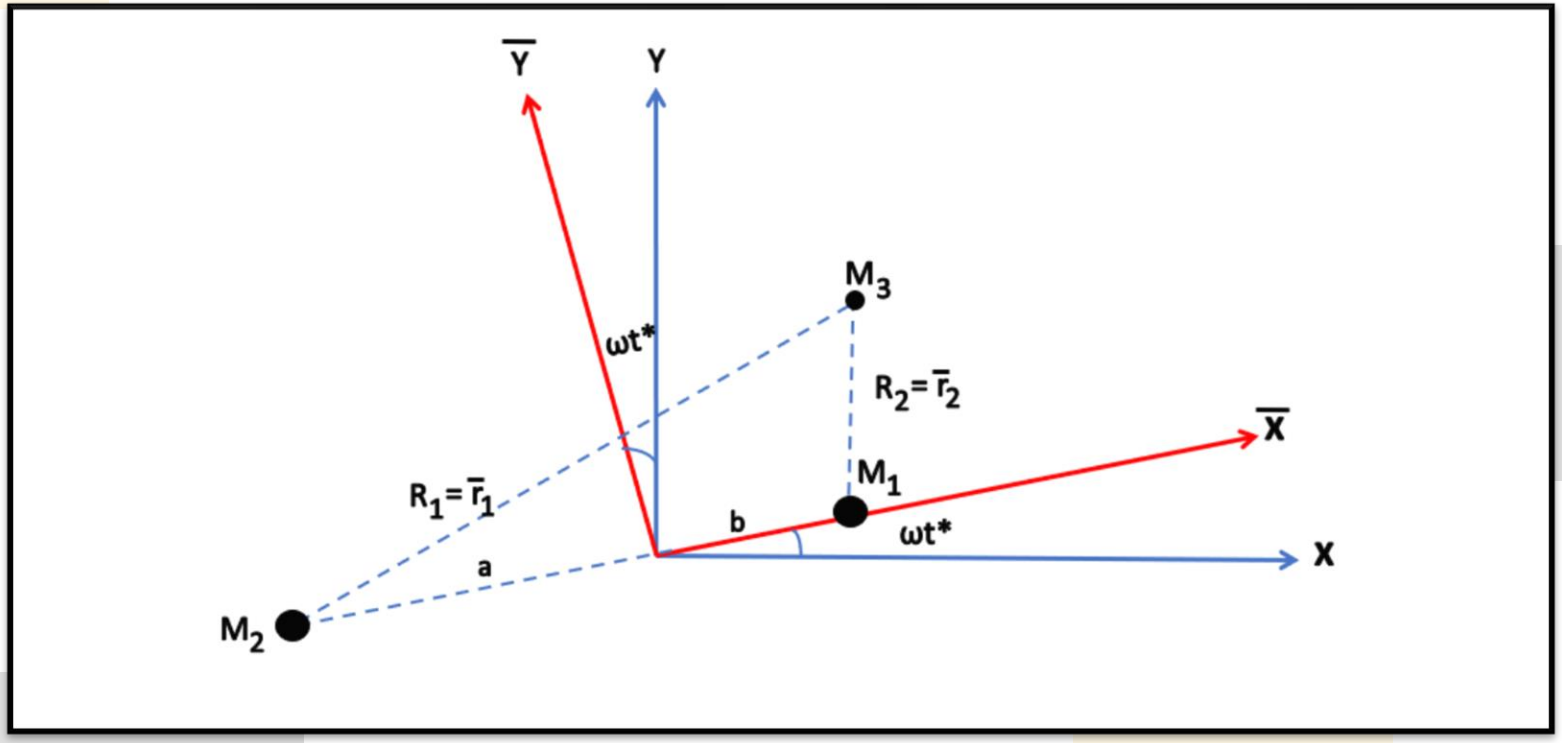



Fig 3. Circular Restricted Three Body Problem

The equation of motion in Circular Restricted Three body problem,

$$\ddot{X} - 2\dot{Y} = \Omega_x$$
$$\ddot{Y} - 2\dot{X} = \Omega_y$$

Where,

$$\Omega = \frac{(\mu_1 r^2{}_1 + \mu_2 r^2{}_2)}{2} + \frac{\mu_1}{r_1} + \frac{\mu_2}{r_2}$$

Jacobi Integral,

$$\left[\dot{X}^2 + \dot{Y}^2\right] = 2\Omega - C$$

# EQUILIBRIUMS POINTS

They are also called as liberation points and Lagrangian points.

$$v = 0$$

Condition,

$$\Omega = C, \Omega_x = 0, \Omega_y = 0$$

$$x - \frac{(1-\mu)(x-\mu)}{{r_1}^3} - \frac{\mu(x+1-\mu)}{{r_2}^3} = \Omega_x = 0$$

$$y(1 - \frac{(1-\mu)}{{r_1}^3} - \frac{\mu}{{r_2}^3}) = \Omega_Y = 0$$

By solving these equations, we get

1. Collinear Liberation points, L1, L2, L3
2. Non- Collinear Liberation points, L4, L5

Where, L1, L2,L3 defined as,

$$-1.271630 \leq x_1 \leq -1$$
$$-1 \leq x_2 \leq 0$$
$$+1 \leq x_3 \leq 1.198406$$

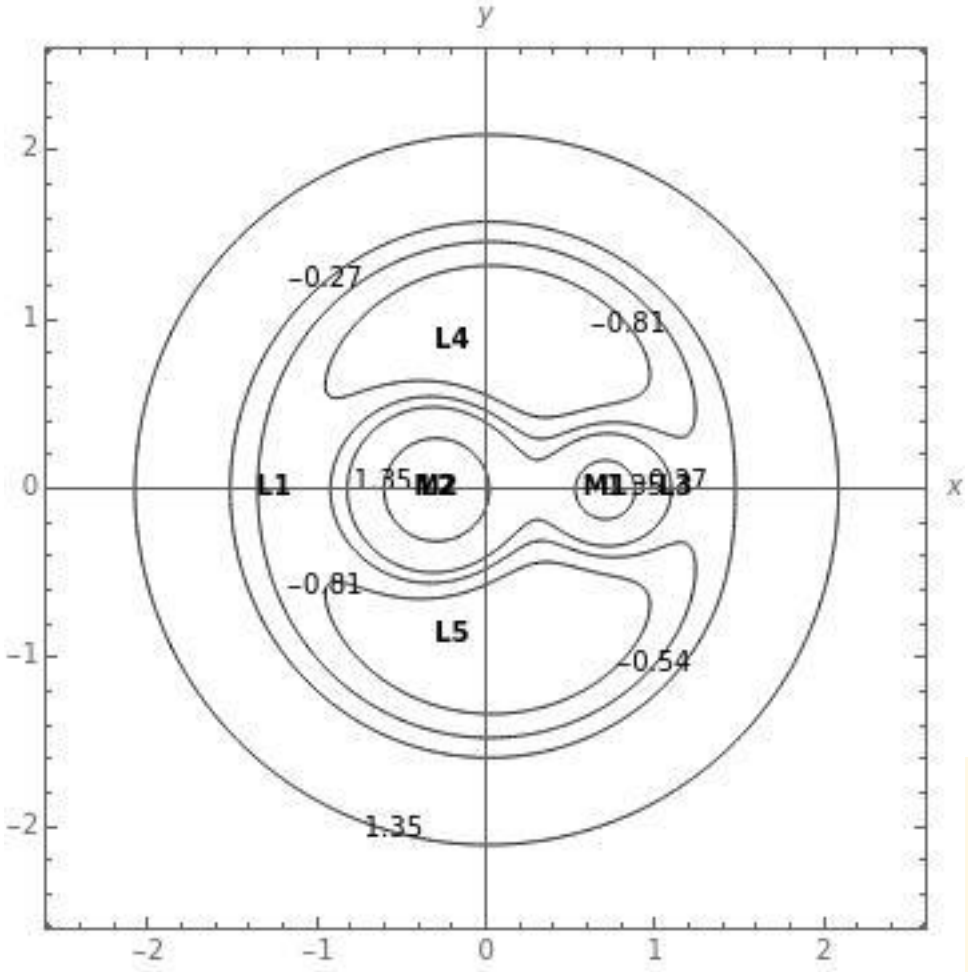
And, L4, L5 as,

$$L_4(\mu - 0.5, 0.866) \text{ and } L_5(\mu - 0.5, -0.866)$$

Where,

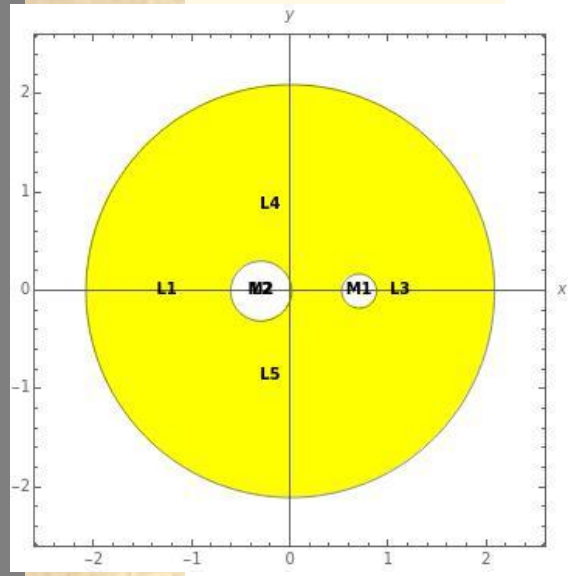
$$0 \leq m_1 = 1 - \mu \leq 0.5$$
$$-1 \leq m_2 = \mu \leq -0.5$$

# ZERO VELOCITY CURVES



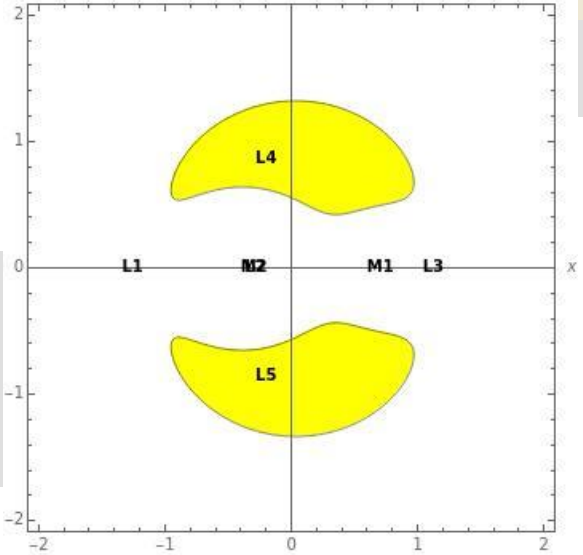

$\mu = 0.3$

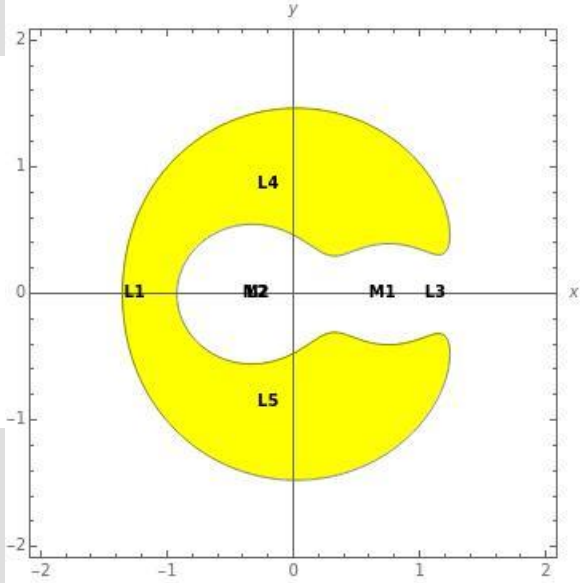
# For $\mu = 0.3$




F = 1.35

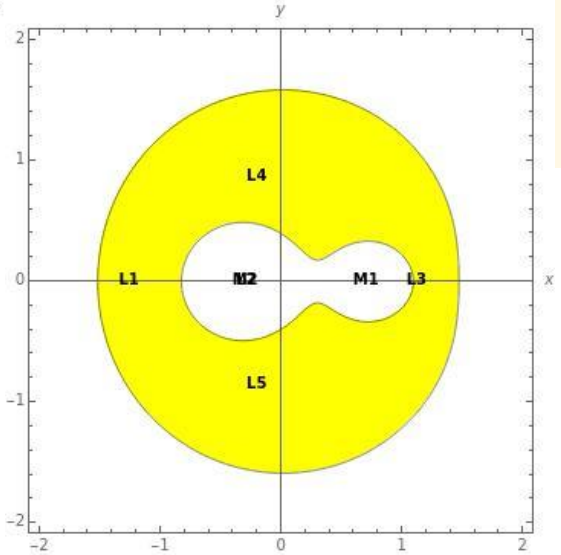
F = -0.81

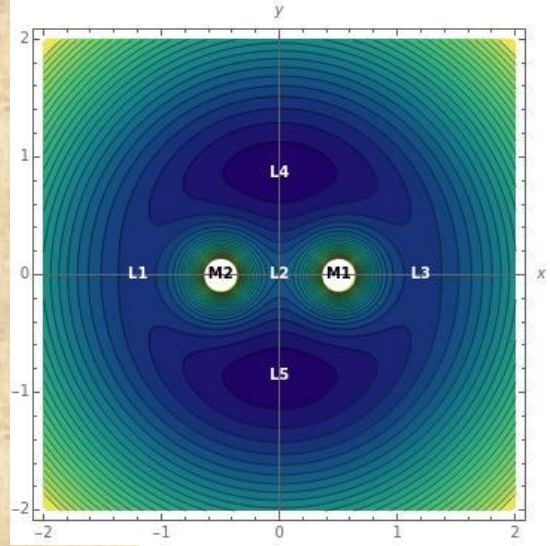



F = -0.54

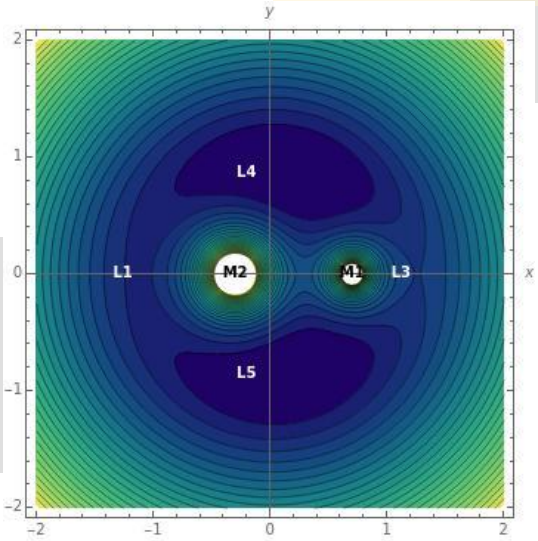
F = -0.27

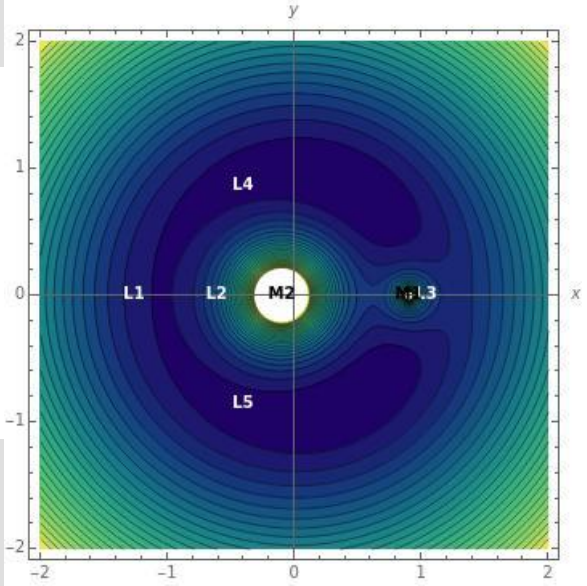





μ = 0.5

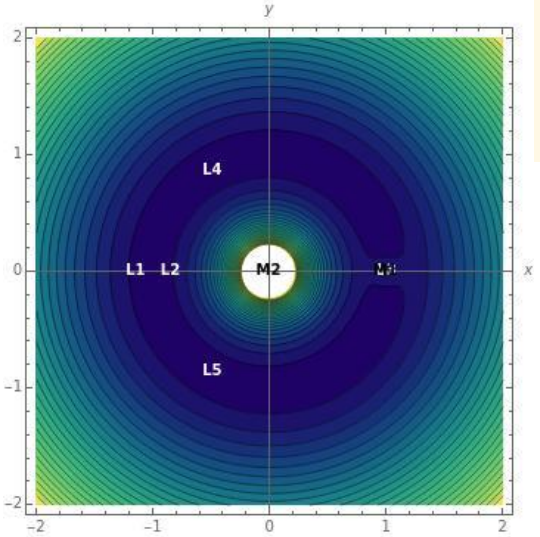
μ = 0.3




μ = 0.1

μ = 0.01



# CONCLUSION & DISCUSSION

# REFERENCES


1. Valtonen, M.J., Karttunen, H., 2006. The three-body problem. Cambridge University Press.

2. Whittaker, E.T., 1917. A Treatise on the Analytical Dynamics of Particles and Rigid Bodies, 2nd.

3. Musielak, Z.E., Quarles, B., 2014. The three-body problem. Rep. Prog. Phys. 77, 065901. https://doi.org/10.1088/0034-4885/77/6/065901

4. Szebehely, V., 2012. Theory of Orbit: The Restricted Problem of Three Bodies. Elsevier.

5. MD Chand Asique, B. S. Rawat, 2023, Mathematical & Computational Study of Three Body Problem in Celestial Mechanics, Internship report.

5. Images has been generated using ppt and Mathematica.

# THANK YOU FOR YOUR TIME